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Experimental study of the mechanisms and defects of single event leakage induced by high-energy heavy ions in AlGaN/GaN diodes

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ABSTRACT

Gallium nitride (GaN) is a promising material for next-generation space microelectronics, demonstrating considerable reliability under low-energy irradiation compared to conventional semiconductors. However, under high-energy irradiation, GaN suffers considerably from single-event effects, where a single heavy-ion strike can cause anomalous behavior such as a permanent increase in leakage current or even the complete destruction of the device. Despite the severity of this issue, an understanding of the physics involved remains limited and has been primarily confined to theoretical simulations. In this study, we investigated the I–V characteristics of AlGaN/GaN diodes affected by single-event leakage current. The analysis shows the increase in forward leakage current originates from a small, localized region in the AlGaN barrier described by a parallel diodes model, while the increase in reverse leakage arises from trap-assisted tunneling in the same damaged region. Transmission electron microscopy imaging confirmed a highly defective AlGaN barrier localized near the anode edge.

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I. INTRODUCTION

With the increasing number of satellites being launched into low-earth orbit, the need for smaller and more power-efficient microelectronics is of growing importance. However, the extreme environment of space presents serious challenges to the reliability of semiconductor technology. One such example is cosmic radiation, which can significantly alter the characteristics of semiconductor devices[1–4] and even cause their complete destruction.[5–7] This has initiated a search for novel semiconductor materials with increased intrinsic radiation resistance for next-generation space electronics. One such candidate is gallium nitride (GaN), which, compared to conventional semiconductors, boasts increased power density,[8] resistance to radiation-induced lattice damage,[9,10] and a 2D electron gas (2DEG).[11–13] This 2DEG is formed via polarization doping,[14] which avoids the radiation-induced compensation from conventional doping techniques employed in other wide-bandgap semiconductors such as SiC and Ga_2O_3. Additionally, because this transport layer is confined to a thin quantum well, the physically small sensitive volume of the channel yields greater resilience to radiation-induced defects.[15,16] These properties have led the AlGaN/GaN high electron mobility transistor (HEMT) to be one of the most active areas of research for new rad-hard electronics. However, the reliability of AlGaN/GaN HEMTs suffers considerably under high-energy irradiation, where a single heavy-ion strike can cause a myriad of phenomena, broadly named single-event effects.

Among these, single-event burnout (SEB) is of special concern. In SEB, a heavy-ion strike interacts with the device's internal electric field to induce catastrophic and irreversible failure often leading to the complete destruction of the device.[17–20] While SEB is a crucial concern for the reliability of GaN HEMTs in space, a comprehensive understanding of the physics involved is lacking. The contemporary computer simulations used to drive understanding of SEB in GaN are typically used solely for qualitative analysis and have poor predictive power, suggesting that the underlying models and assumptions are incomplete. For example,

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defect formation and accumulated damage beyond those already present in pristine devices are not considered.[21,22]

A key concept of SEB is the critical voltage: if the device is biased below the critical voltage during irradiation, it will not experience SEB, and if the device is biased above the critical voltage, it can. Recently, there has been a greater understanding that even below the critical voltage, device damage can still manifest as increased gate or drain leakage, dubbed single-event leakage current (SELC).[19,23] Both SEB and SELC share the same underlying mechanisms: spatially localized charge generation from a heavy-ion strike interacting with the devices' internal electric field.[24,25] The difference is strictly a matter of severity—under SEB the devices experience catastrophic, non-functional damage, while under SELC the devices suffer localized damage but remain functional.[23–25] Since SELC does not destroy the device during testing, this provides a unique opportunity to understand how high-energy heavy-ion irradiation interacts with a biased device to affect its electrical characteristics. Previous studies have identified the gate-drain region of the AlGaN/GaN HEMT to be especially sensitive to SEB/SELC due to its high electric fields.[20,26,27] Post-failure analysis of HEMTs that have experienced SEB typically shows large, localized lattice damage to the active region epi.[28,29] Studies of specifically SELC in GaN HEMTs are sparse, but Kuboyama *et al.* speculated that the increased gate-drain leakage was caused by a local barrier lowering of the Schottky barrier.[30] This was qualitatively shown via simulating a portion of the gate with a zero barrier height. Rasel *et al.* irradiated GaN HEMTs that were both floating and in the on-state and observed in both cases an increase in the leakage current, as well as a lowering of the Schottky barrier height.[31] This was physically attributed to defects at the AlGaN/GaN interface under the gate metal. However, experimental studies of SELC of the lateral GaN HEMT under high stress conditions—off-state, large reverse-bias, and high energy heavy-ions—remain understudied.

To investigate the physics of the gate-drain HEMT region without the increased complexity of a three-terminal device, we employ the lateral AlGaN/GaN Schottky diode as our model device. While heavy-ion effects in GaN diodes have previously been explored, these studies have focused predominantly on the p-i-n architecture.[19,32,33] The heterostructure's polarization field as well as a surface-dominated lateral electric field yields a vastly different electrostatic profile, necessitating separate analysis. We irradiated AlGaN/GaN Schottky diodes with 1713 MeV Au ions (8.7 MeV/u, LET of 69.3 MeV cm^2/mg) under bias to systematically investigate the mechanics governing heavy-ion induced leakage in GaN devices.

II. EXPERIMENT DESIGN AND METHODS

A. Device fabrication

Device fabrication started with pGaN/UID GaN/$Al_{0.27}Ga_{0.73N}$/UID GaN epitaxy grown via metal-organic chemical vapor deposition on C-doped semi-insulating GaN substrate from Mitsubishi Chemical Corporation. GaN-on-GaN has a minimized defect density due to a lack of lattice and thermal expansion mismatch,[34] which are inherently present in the conventional heteroepitaxial growth on silicon and sapphire.[35,36] This helps to reduce the effects of pre-existing defects on our study. This epi stack was previously used to study the effects of heavy ions on a RESURF HEMT structure,[37] so the GaN cap needed to be removed in order to match the epitaxial stack with conventional devices. This was achieved via selective removal using Cl_2/O_2 inductively coupled plasma reactive ion etching (ICP-RIE) and was confirmed to be completely removed using a profilometer. Next, isolation mesas were formed using BCl_3/Cl_2 ICP-RIE. Then, e-beam evaporated Ti/Al/Ni/Au ohmic contacts for the cathode were deposited followed by an 850 °C anneal in the N_2 atmosphere. The anode was formed via Ni/Au e-beam evaporation and a 550 °C anneal in O_2. Finally, 100 nm of Si_3N_4 passivation was deposited with plasma-enhanced chemical vapor deposition followed by a CHF_3 ICP-RIE to open the pads for probing. The final device structure is shown in Fig. 1.

Chips were then diced and characterized. The SMU had a nominal noise floor of ∼10 pA for the I–V sweeps. A nominal breakdown voltage of −600 V was identified. Next, they were packaged into TO-3 packaging to interface with the existing radiation testing setup and then underwent heavy-ion irradiation testing.

B. Radiation testing

The heavy-ion testing was performed at the Texas A&M Cyclotron Institute using Au ions with an energy of 1713MeV (8.7 MeV/u, LET of 69.3 MeV cm^2/mg). This corresponds to an ion range of 48 μm, orders of magnitude larger than the active region thickness. The samples were uniformly irradiated at a normal incidence with a nominal flux of 1×10^4 ions/cm^2 s. The beam energy and flux values were chosen to be in accordance with existing standards.[38,39] More details on the heavy-ion testing setup can be found in the supplementary material of our previous related study.[37]

The applied voltage during irradiation must be carefully chosen to observe SELC effects. If the applied voltage is too low, the field may not be high enough to induce any single-event effects. Inversely, if the applied voltage is too high, single-event burnout will occur, destroying the device and rendering electrical characterization impossible. For this experiment, we first used a sacrificial device to identify the critical voltage at which single-event burnout occurs. Then, the remaining devices were biased below the critical voltage at a range of biases to ensure SELC occurs while single-event burnout does not.

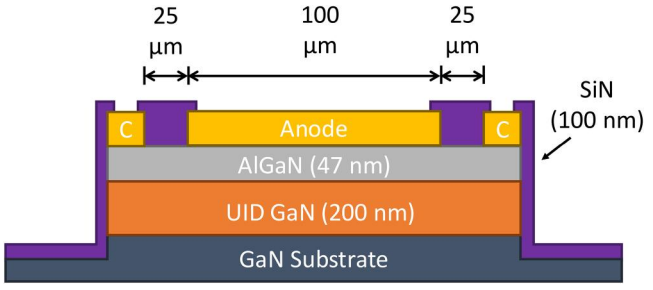


FIG. 1. The AlGaN/GaN diode structure used in this study.

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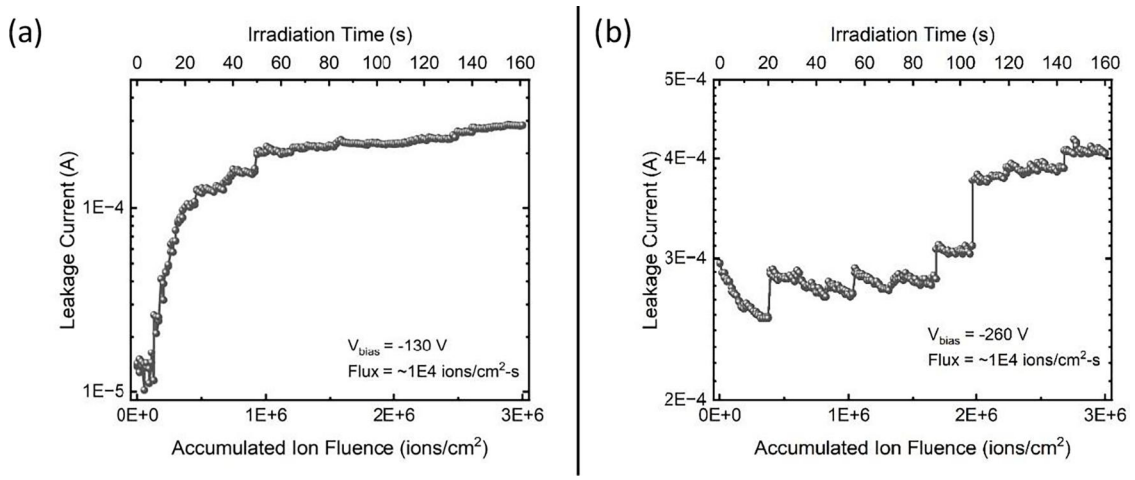


FIG. 2. Captured leakage transients during irradiation for (a) the diode biased at −130 V and (b) the diode biased at −260 V.

First, a single-event characterization test was performed on one of the diodes to identify the SEB critical voltage. The diode was initially biased at −150 V and irradiated at a lower fluence (3×10^5 ions/cm^2) than subsequent experiments to efficiently determine the critical SEB threshold voltage. Since the device survived, the reverse voltage was increased by an additional 50 to −200 V and the same device was irradiated again with the same fluence. This process was repeated until the sacrificial diode failed catastrophically at a voltage of −300 V, the critical SEB voltage of our devices. Since the non-irradiated diodes failed at −600 V, this suggests a derating factor of ∼50%, nominally comparable to previously reported results for GaN HEMTs.[17,28] Post-failure analysis using scanning-electron microscopy revealed significant surface damage localized on the anode edge (Fig. S1 in the supplementary material), analogous to the gate-drain damage observed in prior literature.[17,26,28] Beyond this, the sole purpose of this diode was to determine the critical SEB voltage, and no data from this device are used in our subsequent analysis.

Therefore, to ensure the remaining diodes would survive testing, the following irradiation had the diodes biased below the critical voltage of the sacrificial device. Because the threshold voltage for inducing SELC was unknown, two bias conditions, −130 and −260 V, were chosen. Alongside the biased diodes were multiple floating diodes which served as controls in this study. Using floating diodes as controls enables the separation of effects caused solely by irradiation (e.g., beam heating, displacement damage, and ionization damage) from those caused by irradiation in conjunction with electrical bias, since SELC and SEB require an applied device bias to occur.[24]

All devices were irradiated to a fluence of 3×10^6 ions/cm^2. During irradiation, the current transients of the biased devices were monitored, as shown in Fig. 2 as a function of time and the accumulated ion fluence.

In Fig. 2(a), the increase in leakage current as a function of the irradiation time is clear, consistent with the SELC signature observed in SiC MOSFETs.[23] In Fig. 2(b), the leakage current also increases as a function of the irradiation time. In particular, the characteristic "jumps" in leakage current are more clearly observed in the −260 V diode, likely due to the higher reverse bias voltage amplifying the maximum electric field and making individual ion strikes more impactful.[24]

While Fig. 2(a) shows the leakage current initially starting at the SMU noise floor, the Fig. 2(b) measurement begins relatively high at 3×10^{-4} A. We believe this is because during irradiation the higher reverse bias triggered an immediate increase in the leakage current faster than the SMU sampling rate and not due to sample damage during transit/preparation. Right before irradiation, we did a pre-irradiation test of the forward and reverse characteristics to ensure the devices behave as expected. We observed no discrepancy between measurements at the cyclotron and that in the home lab for all our biased devices, and the reverse leakage was at or below the noise floor for all of them before the beam turned on.

After irradiation was complete, the samples left the cyclotron facility to undergo more *ex situ* measurements for comparison with the pre-radiation characteristics.

III. RESULTS AND DISCUSSION

The I–V characteristics of various diodes are shown in Fig. 3. All diodes had their series resistance nominally unaffected by irradiation. For the diodes that were floating during irradiation, the forward characteristics remained relatively unchanged, while the reverse leakage current increased about one order of magnitude at −10 V [Fig. 3(a)]. In comparison, the diodes that were biased during irradiation had significantly different forward characteristics below the knee voltage, and in reverse had approximately four orders of magnitude current increase at −10 V [Figs. 3(b) and 3(c)]. Analysis of these diodes is split between the forward and reverse characteristics.

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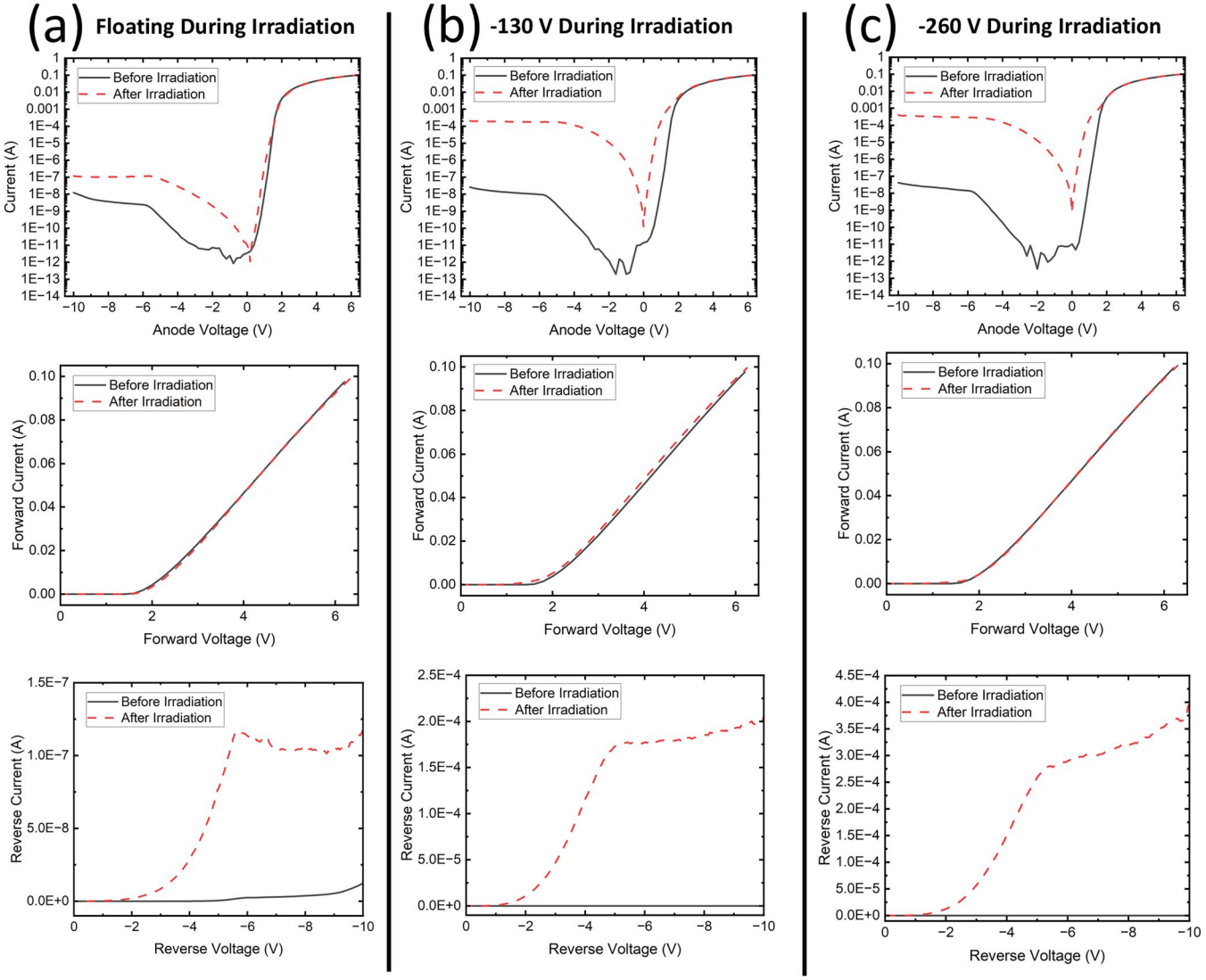


FIG. 3. Semi-log and linear I–V characteristics before and after irradiation of (a) a diode that was floating during irradiation, (b) a diode that was biased at −130 V during irradiation, and (c) a diode that was biased at −260 V during irradiation. The radiation was Au at 1713 MeV with a flux of 1×10^4 ions/cm^2 s and a fluence of 3×10^6 ions/cm^2. The SMU had a nominal noise floor of ∼10 pA for the I–V sweeps.

A. Forward characteristics

As seen in the semi-log plots of Fig. 3, the diodes that were biased during irradiation have a significantly higher forward current below ∼1.5 V compared to the floating diodes. To better understand this phenomenon, the classic thermionic emission model is used as a starting point. This is modeled by the following equations:[40]

$$J = J_s\left(e^{\frac{q(V-JR_s)}{nkT}} - 1\right), \tag{1}$$

$$J_s = A^* T^2 e^{-\frac{q\phi_b}{kT}}, \tag{2}$$

where J_s is the saturation current density, q is the elementary charge, V is the applied voltage, R_s is the area specific resistance, n is the ideality factor, k is the Boltzmann constant, T is the temperature, A^* is the Richardson constant, and ϕ_b is the Schottky barrier height. While the floating diodes after irradiation were well described by Eq. (1) [Fig. 4(a)], the biased diodes were no longer well described by an ideal Schottky diode [Figs. 4(b) and 4(c)] as shown by the lower than expected current in the 1–3 V range. Additionally, a small bump is observed at approximately 1.5 V in the biased diodes, which is not predicted by the conventional model.

According to existing single-event effects theory,[19,23] a heavy-ion striking a device in a region with a high internal electric

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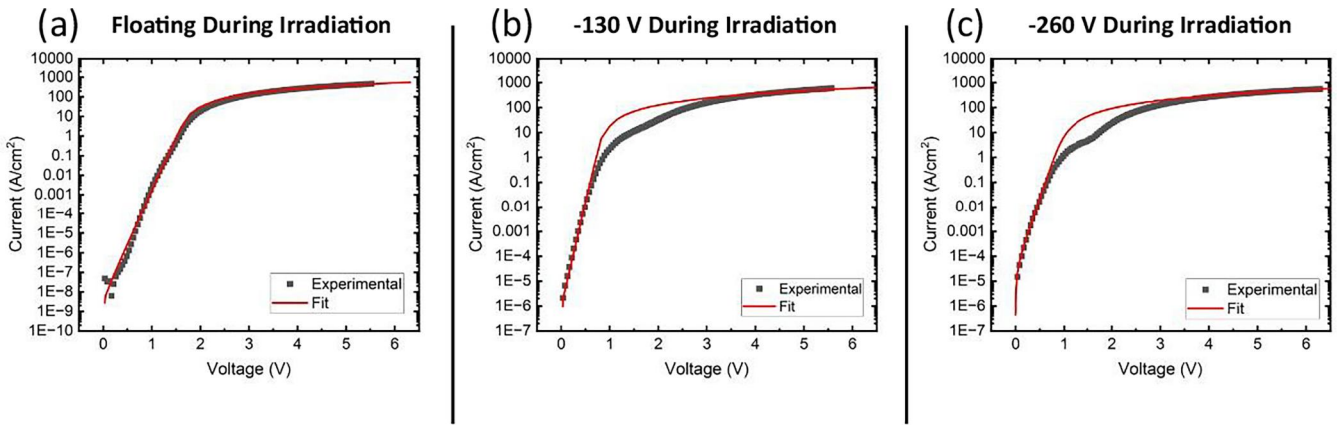


FIG. 4. Forward J–V characteristics after irradiation of (a) a floating diode, (b) the diode irradiated at −130 V, and (c) the diode irradiated −260 V. The solid red line is the best fit according to the conventional Schottky diode model.

field can induce Joule heating, creating localized damage. In the context of this theory, we propose two key characteristics of the damaged region to better explain the forward characteristics of the AlGaN/GaN diodes affected by SELC: (1) a lowered Schottky barrier height and (2) the region volume is very small compared to the entirety of the device. From these assumptions, we can model the behavior of our AlGaN/GaN diodes affected by SELC as two diodes in parallel (Fig. 5). One diode (original) represents the characteristics of the undamaged region of the diode, and the other diode (barrier lowered or BL) represents the characteristics of the heavy-ion damaged region. While existing theory predicts that the resistivity of the BL diode will be reduced, the total resistance of the BL diode can be much higher than the resistance of the original diode by virtue of the BL diode having a smaller

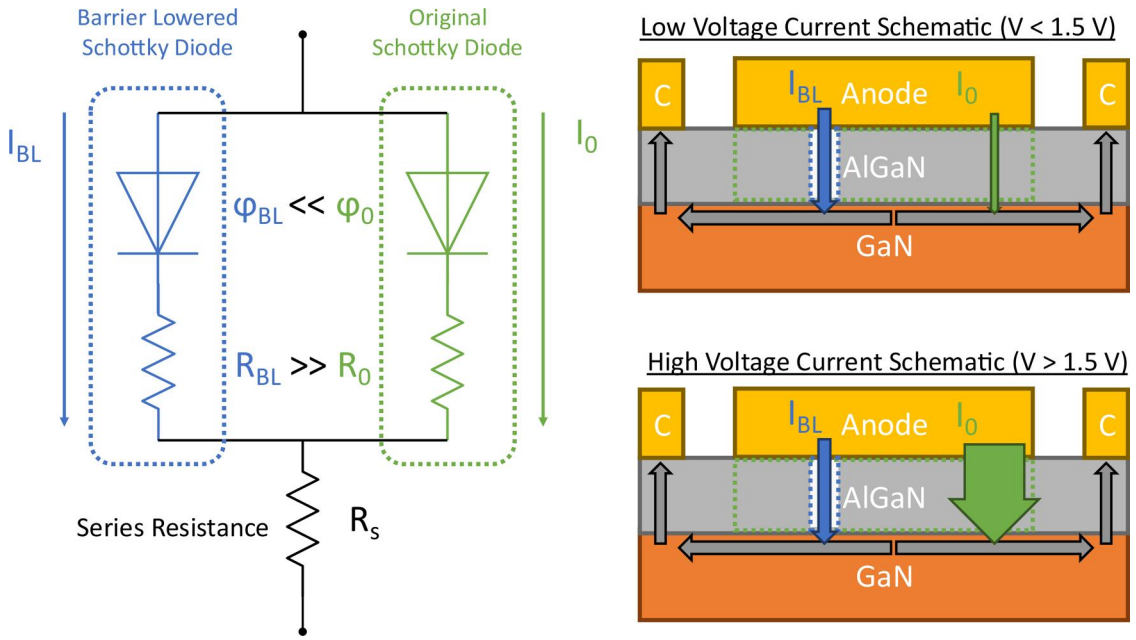


FIG. 5. Equivalent circuit and physical representation of the parallel diodes model proposed in this paper. The device's internal electric fields in conjunction with a heavy-ion strike create localized regions with lowered barrier heights. This causes the devices affected by single-event leakage to behave like two Schottky diodes with different barrier heights and internal resistances in parallel.

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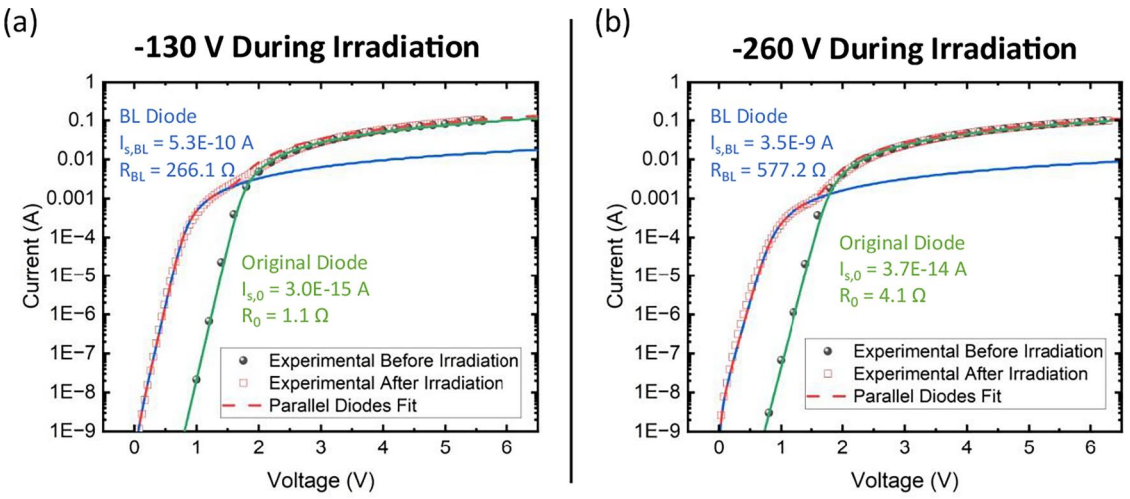


FIG. 6. Experimental fit to the parallel diodes model for (a) the diode irradiated at −130 V and (b) the diode irradiated at −260 V. The solid blue line is a fit to the radiation-induced barrier lowered diode, while the solid green line is a fit to the pre-irradiation, original diode. The dashed red line represents the total fit according to the parallel diodes model. While we can extract the reverse bias saturation current of the BL diode ($I_{s,BL}$), we cannot extract the true barrier height because the area of the damaged region is unknown.

volume. This model is similar to one developed in SiC Schottky diodes, where the same observation of a double barrier in the I–V characteristics is modeled by two parallel diodes with different resistances and barrier heights.[41–43] In SiC, this effect is attributed to defects in the epi beneath the anode, which agrees with our later imaging results.

From an unnormalized form of Eq. (1) and circuit analysis of Fig. 5, the following equations can be derived:

$$I_0 = I_{s,0}\left(e^{\frac{q(V-IR_S-I_0R_0)}{n_0kT}} - 1\right), \tag{3}$$

$$I_{BL} = I_{s,BL}\left(e^{\frac{q(V-IR_S-I_{BL}R_{BL})}{n_{BL}kT}} - 1\right), \tag{4}$$

$$I = I_0 + I_{BL}\,. \tag{5}$$

For qualitative analysis, there are two major regimes that are described by this equation—low and high total current. At a low total current, since ϕ_0 is larger than ϕ_{BL}, the original diode contributes negligibly to the total current. Therefore,

$$I \sim I_{s,BL}\left(e^{\frac{q(V-I[R_S+R_{BL}])}{n_{BL}kT}} - 1\right) \tag{6}$$

For a high total current, since we assumed that the heavy-ion damaged region is much smaller than the total device volume, it is reasonable to assume that $R_{BL} \gg R_0$. So the $I_{BL}R_{BL}$ voltage drop is significant, and the barrier lowered diode has a minor effect,

$$I \sim I_{s,0}\left(e^{\frac{q(V-I[R_S+R_0])}{n_0kT}} - 1\right). \tag{7}$$

Figure 6 shows the experimental data fitted to the parallel diodes model. The original diode characteristics can be observed after irradiation in the high current regime but is masked by the BL diode at low currents. Also, the parallel diodes model provides a satisfying explanation for the presence of the bump in the current at ∼1.5 V—it is caused by the original diode turning on while the BL diode is in its $I_{BL}R_{BL}$ dominated region.

Since the area of the heavy-ion damaged region is unknown, we cannot extract the BL diode's true Schottky barrier height from Eq. (2). Instead, an apparent Schottky barrier height can be calculated using the total device area for normalization. Although this does not represent the local barrier height of the damaged region, this enables us to better evaluate how the BL diode affects the overall device characteristics.

The forward current–voltage characteristics of the diodes at various temperatures were measured before and after irradiation. A J–V–T profile of the diode irradiated at −130 V is shown in Fig. 7. Taking the natural logarithm of Eq. (1) and assuming the exponential term is much greater than one and the current density is sufficiently small yield

$$\ln(J) = \ln(J_s) + \frac{qV}{nkT}. \tag{8}$$

By performing a fit in the linear region of the semi-log J–V curve, the barrier height and the ideality factor can be calculated from the extracted intercept and slope, respectively. Examples of the

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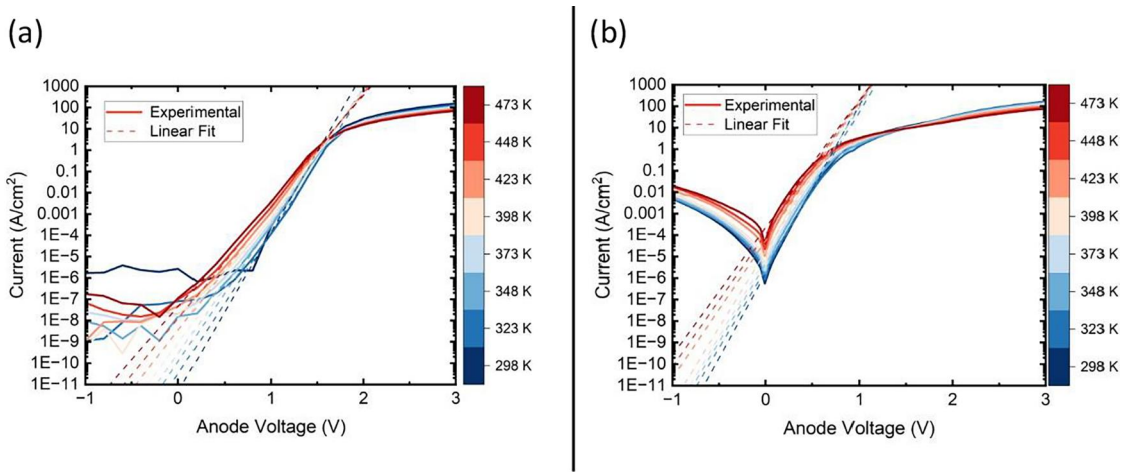


FIG. 7. A forward J–V–T profile of the −130 V device (a) before and (b) after irradiation. The solid line shows the experimental data, while the dashed line shows the linear fit used for extraction of the Schottky barrier height and ideality factor. (b) is normalized against the total device area rather than the area affected by heavy-ion damage as the latter is unknown.

linear fits are shown as the dashed lines in Fig. 7, and the calculated Schottky barrier heights and ideality factors extracted from Fig. 7 for each temperature are given in Fig. 8. The barrier height was calculated using a Richardson constant of 30.6 $A/cm^2 K^2$, determined by applying Vegard's Law to the effective electron masses of GaN[44] and AlN.[45] As seen in Fig. 8, the apparent barrier height after irradiation has been lowered, but no conclusions can be drawn regarding the ideality factor—when this analysis was performed on the diode irradiated at −260 V, the ideality factor increased after irradiation in contrast to the decrease for the −130 V diode.

In an ideal Schottky diode, the barrier height is temperature independent and the ideality factor equals one. However, Fig. 8 shows that for both pre- and post-irradiation, the barrier height increases with temperature, while the ideality factor exceeds one and decreases as the temperature rises. This behavior has previously been observed across a wide range of semiconductors and has been attributed to spatial barrier height inhomogeneities.[46,47] A model developed by Werner and Güttler[47] assumes a Gaussian distribution of the barrier heights, resulting in the apparent Schottky barrier height being described by

$$\phi_b = \overline{\phi_b} - \frac{q\sigma^2}{2kT}, \tag{9}$$

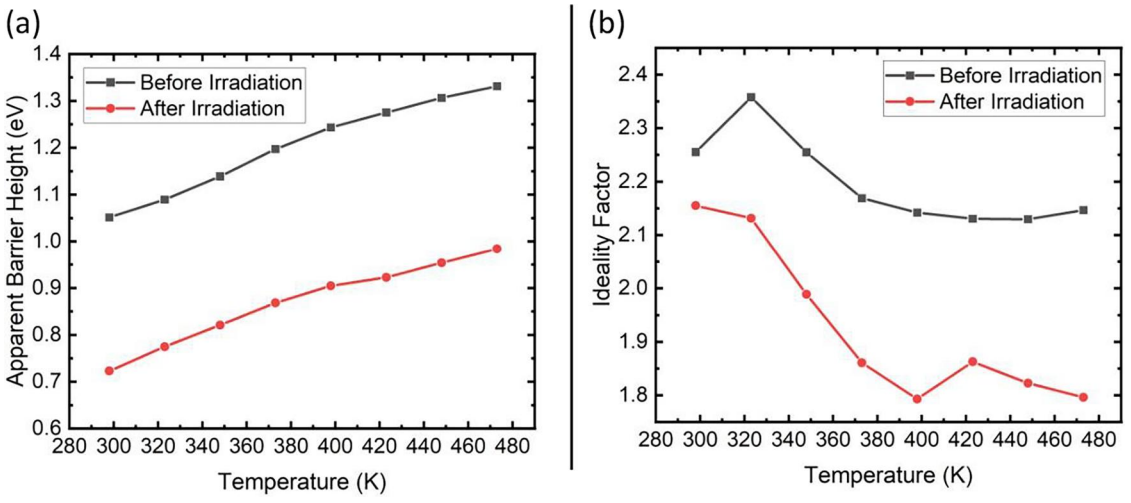


FIG. 8. Extracted (a) apparent Schottky barrier height and (b) ideality factor from Fig. 7 across multiple temperatures before and after irradiation.

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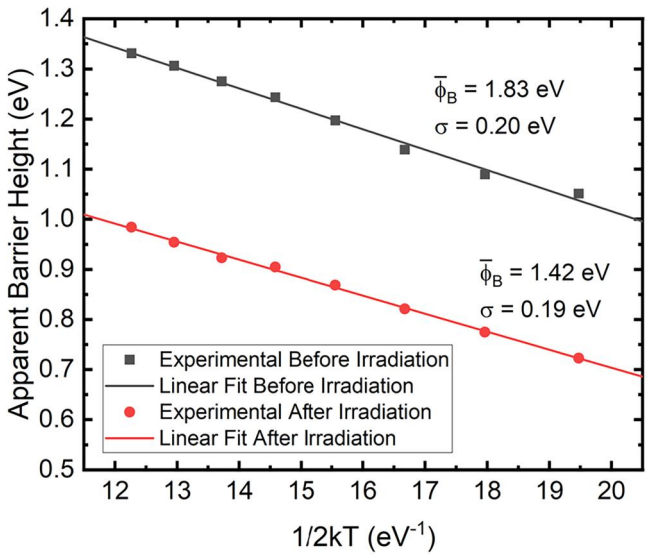


FIG. 9. Apparent Schottky barrier height vs 1/2 kT. Assuming a Gaussian distribution of barrier heights, the mean barrier height and standard deviation can be derived from the linear fit, shown as the solid lines.

where $\overline{\phi_b}$ is the mean barrier height and σ is the standard deviation of the barrier height. Therefore, $\overline{\phi_b}$ and σ can be extracted from a linear fit of the measured ϕ_b vs q/2kT, as shown in Fig. 9.

Substituting Eq. (9) into Eqs. (2) and (8) yields Eq. (10), showing that a modified Richardson plot can be used to verify the validity of our assumptions, as seen in Fig. 10. The good linearity of the plot shows the fabricated devices' behavior agrees with the expected results for an inhomogeneous Schottky diode,

$$\ln\left(\frac{J}{T^2}\right) - \frac{q^2\sigma^2}{2k^2T^2} = \ln(A^*) - \frac{q\bar{\phi}_b}{kT}. \tag{10}$$

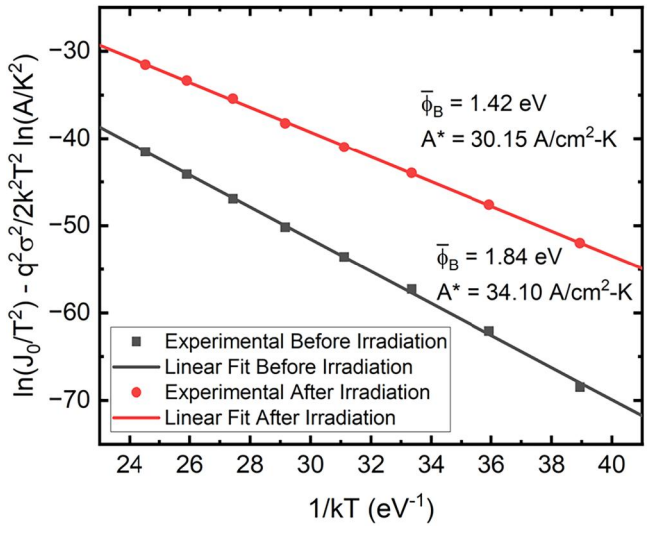


FIG. 10. Modified Richardson plot before and after irradiation.

This characterization technique was also applied to the diode irradiated at −230 V. A summary of the results is shown in Table I. Both diodes showed a decrease in their apparent mean Schottky barrier height after irradiation, while their standard deviation of the barrier height remained relatively unchanged. The extracted A* is included for completeness; however, it should be interpreted with caution. Since A* is calculated by extrapolating over a long distance to 1/kT = 0, along with the fact that it is derived from an exponential function, the uncertainty in A* is not negligible.[40] The series resistance of both diodes remained nominally unchanged after irradiation. This is in good agreement with the previous literature and is attributed to the small sensitive volume of the 2DEG drift region, which minimizes the likelihood of radiation-induced defect formation within this volume.[15,16]

It is important to emphasize that the post-radiation analysis measures only the apparent Schottky barrier height of the diode and not the actual barrier height across the entire device area. In the framework of the parallel diodes model, a small, localized region with significantly increased leakage dominates the low voltage response of the entire device. This can give the appearance that the barrier height of the entire diode has been lowered, when in reality the effect is only in a small region. This region will also have significant consequences for the reverse-bias characteristics, as shown in Sec. III B.

B. Reverse characteristics

As shown in Fig. 3, the reverse leakage of the AlGaN/GaN diodes biased during irradiation is orders of magnitude higher than that of the floating devices. According to the proposed parallel diodes model, the increased leakage is a localized effect caused by the heavy-ion damaged region rather than being uniformly distributed across the entire diode area. As in Sec. III A, the reverse leakage mechanism will be analyzed on the apparent current density normalized across the entire device area. However, this will not affect the conclusions of our analysis, as described later.

Typically, the reverse leakage current of an AlGaN/GaN diode is attributed to a Poole–Frenkel process[48,49] as described by[50]

$$J_{PF} = CEe^{-q\phi_t/kT+\sqrt{E}\sqrt{q/\pi\epsilon_d\epsilon_0}/kT}, \tag{11}$$

where C is a constant, E is the electric field across the AlGaN barrier, ϕ_t is the trap level below the conduction band, ε_d is the relative dynamic permittivity of the AlGaN, and ε_0 is the vacuum permittivity. From charge neutrality and Gauss's law, the electric field in the AlGaN barrier for an AlGaN/GaN heterostructure can be estimated by[48]

$$E(V) \sim \frac{\sigma_p - qn_{2DEG}(V)}{\epsilon_r\epsilon_0}, \tag{12}$$

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TABLE I. Apparent mean Schottky barrier height, standard deviation, Richardson constant, and series resistance before and after irradiation.

	$\overline{\phi_b}$ (eV)		σ (eV)		A* (A/cm^2 K)		R_s (Ω mm)	
	Before	After	Before	After	Before	After	Before	After
Irradiated at −130 V	1.84	1.42	0.20	0.19	34.10	30.15	19.66	19.37
Irradiated at −260 V	1.87	1.55	0.21	0.21	32.00	32.59	19.66	19.38

where σ_p is the fixed polarization charge, n_{2DEG} is the electron sheet density at the AlGaN/GaN interface, and ε_r is the relative static permittivity of the AlGaN. n_{2DEG} as a function of the applied voltage can be extracted from integrating the capacitance–voltage curve [Fig. 11(a)]. While $Al_{0.27}Ga_{0.63}N$ has a theoretical value of 1.5×10^{13} cm^{-2} for σ_p,[14] in practice the experimental value is typically much smaller. Given that the measured n_{2DEG} at zero bias was $\sim 6.5 \times 10^{12}$ cm^{-2} compared to the theoretical value of 1.4×10^{13} cm^{-2}, σ_p was adjusted to 0.8×10^{13} cm^{-2} to match the experimental 2DEG density. Using this polarization charge allows for the derivation of Fig. 11(b) using Figs. 11(a) and Eq. (12).

Equation (11) predicts a linear dependence of ln(J/E) vs $\sqrt{E}$ for a Poole–Frenkel dominated conduction process. Figure 12 shows this relationship for the irradiated diodes. As can be seen in Fig. 12(a), the diode that was floating during irradiation has a linear ln(J/E) vs $\sqrt{E}$ plot, consistent with a Poole–Frenkel process. However, for the diodes biased during irradiation [Figs. 12(b) and 12(c)], the plot is highly non-linear, implying a leakage mechanism separate from Poole–Frenkel.

It should be noted that although the current density used in Figs. 12(b) and 12(c) is normalized by the entire device area rather than the damaged area, it will not affect the analysis—the area used for normalization only translates the plot vertically and does not change the linearity of the plot.

Another possible leakage mechanism that has been observed in AlGaN/GaN diodes is trap-assisted tunneling (TAT). Traps in the AlGaN barrier can enhance the tunneling probabilities of electrons, increasing the leakage current. This process is described by[51]

$$J_t \sim e^{-\frac{4\sqrt{2qm_{AlGaN}}}{3\hbar}\Phi_t^{\frac{3}{2}}/E}, \tag{13}$$

where m_{AlGaN} is the effective mass of the electron in the AlGaN barrier and Φ_t is the trap level with respect to the conduction band. From Eq. (13), it can be observed that the plot of ln(J) vs 1/E is linear if TAT is a dominant leakage process. Like with Poole–Frenkel, using the total diode area for normalization does not affect the plot linearity or slope. As can be seen in Fig. 13, the reverse leakage of the diodes irradiated at −130 and −260 V is well described by TAT. From the slope of the TAT plot, the trap level can be extracted (Fig. 14). A value of $0.25m_0$ was used for m_{AlGaN} calculated from Vegard's law.

Over the temperature range of 298–473 K, the extracted trap levels were between 0.21 and 0.27 eV. For both the −130 and −260 V devices, the extracted trap levels at each temperature were in good agreement, suggesting that the damage mechanism in both samples is identical. The extracted trap energies also showed a decreasing trend with temperature. This phenomenon has been

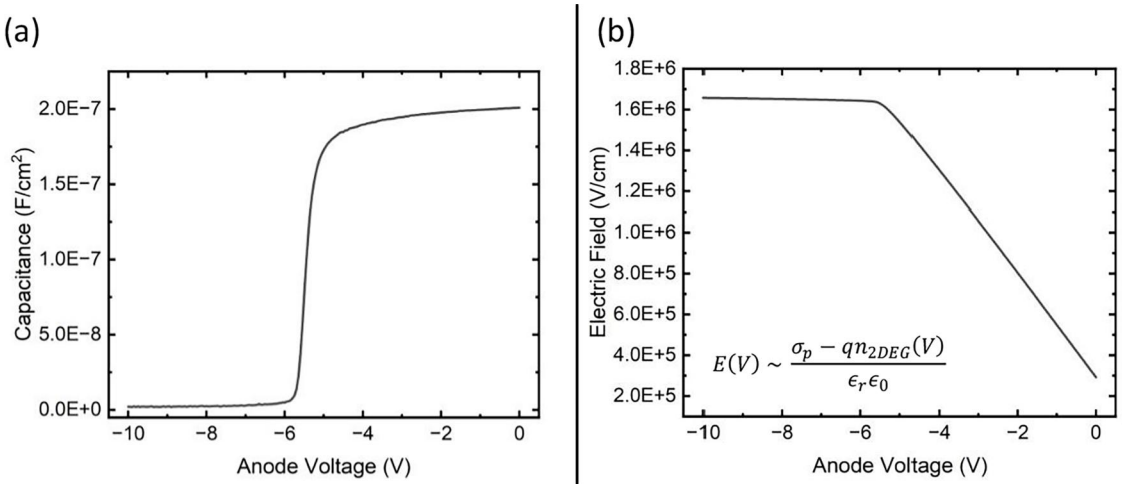


FIG. 11. (a) The capacitance curve and (b) the calculated surface electric field of the −130 V irradiated device.

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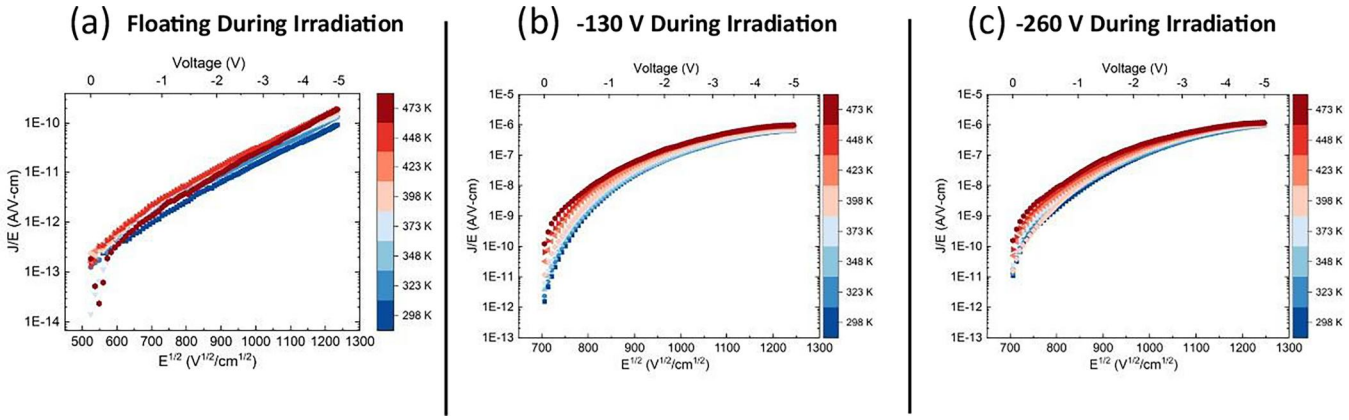


FIG. 12. The J/E vs $\sqrt{E}$ semi-log plots of (a) a diode that was floating after irradiation (b) the diode that was biased at −130 V during irradiation and (c) the diode that was biased at −260 V during irradiation. The non-linearity of (b) and (c) suggests a reverse leakage mechanism that is not Poole–Frenkel.

previously observed in AlGaN/GaN transistors[52,53] and has been attributed to the thermal activation of electrons.[52]

C. Defect identification

To better understand the physical manifestation of SELC damage in our devices, we performed cross-sectional bright-field transmission electron microscopy (TEM) analysis on both the floating sample and the −130 V biased sample, as shown in Fig. 15. All samples experienced identical lamella preparation. Figures 15(a) and 15(c) are the floating sample and the −130 V sample imaged near the anode edge, while Fig. 15(b) is the −130 V sample imaged toward the anode center. While we intended to image the exact anode edge where the highest damage should theoretically occur, slight drift during lamella thinning shifted the imaging window for the floating sample to the anode side, while the biased sample window was in the access region.

A clear contrast between the cross sections is evident. In the floating sample [Fig. 15(a)] and in the biased sample near the anode center [Fig. 15(b)], the AlGaN barrier remains pristine, while in the biased sample near the anode edge [Fig. 15(c)] the AlGaN barrier is highly damaged. This finding provides a validation of our developed model and its consistency with existing single-event theory. Away from the high-field edge, the AlGaN barrier remains as pristine as the control device. The severe damage of the AlGaN barrier is strictly localized in the regions with the highest internal electric field. Notably, these defects are confined to the AlGaN barrier and are absent in the GaN buffer

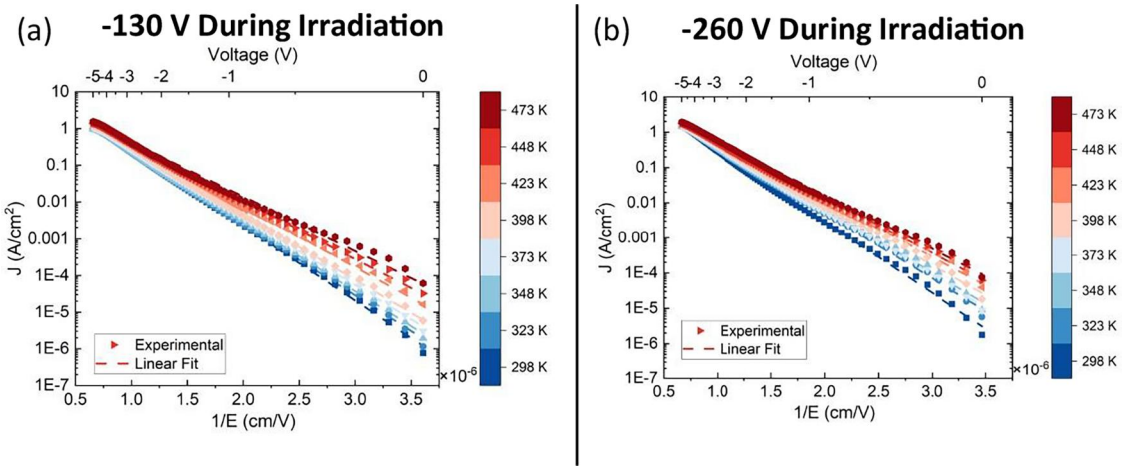


FIG. 13. The semi-log J vs 1/E plots of (a) the diode that was biased at −130 V during irradiation and (b) the diode that was biased at −260 V during irradiation. The good linearity of the plots suggests a trap-assisted leakage mechanism.

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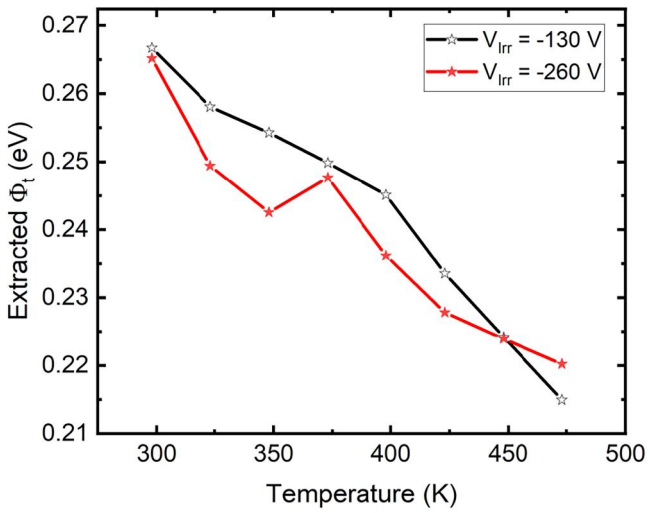


FIG. 14. Extracted trap level from the slope of ln(J) vs 1/E across various temperatures for the −130 and −260 V diodes. The decrease in trap levels with increasing temperatures has been attributed to the thermal activation of electrons.[52]

layer. This distribution is expected, as these regions experience the highest current density and electric field, and consequently the maximum Joule heating.

Previous studies of SELC in GaN p–i–n diodes have attributed the increase in device leakage to conductive micropipette formation from heavy-ion tracks,[19] but no micropipettes are observed in Fig. 15(c). Rather, the defect formation appears to be clusters distributed within the AlGaN barrier. We propose that these defects are formed not from the latent track of a heavy ion, but rather from Joule heating after a heavy-ion strike induces a large current pulse. Since the anode edge has the highest internal electric field, it yields the greatest power dissipation, causing the defects to form there preferentially. We speculate that improved electric field and thermal management of this region can lead to an improved SEE response.

IV. CONCLUSION

To summarize, the forward and reverse characteristics of AlGaN/GaN diodes affected by SELC have been reported. The forward I–V curves suggest that the increased leakage current induced by SELC is not uniformly distributed throughout the entire diode area, but rather originates from a small, localized damaged region. This behavior can be explained by a proposed

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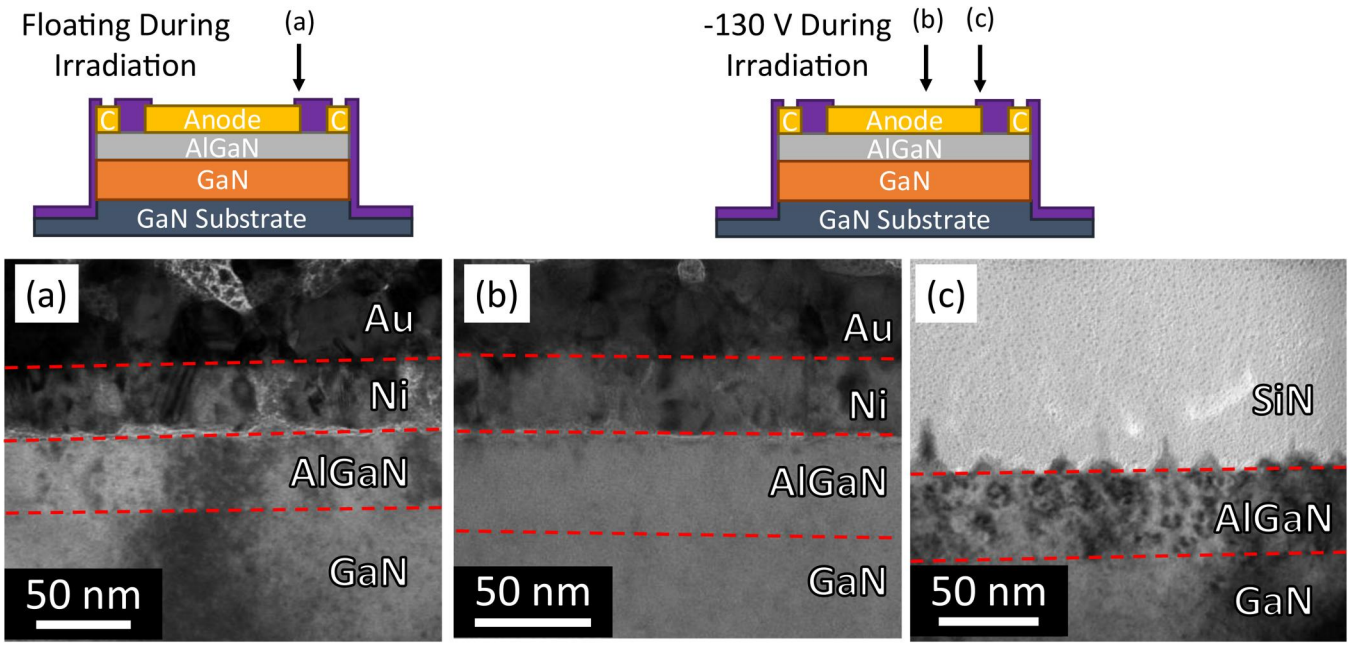


FIG. 15. Cross-sectional TEM images of (a) the floating device and [(b) and (c)] the device biased at −130 V during irradiation. (a) and (c) were taken near the anode edge, while (b) was taken more toward the center of the anode. Although imaging of (a) and (c) was intended to occur on the exact pad edge, slight deviations during lamella preparation resulted in the imaging window for (a) to be under the anode, while (c) was within the access region. It should be noted that surface roughness/spikes on the AlGaN surface are artifacts of the fabrication etch step and are unrelated to irradiation damage.

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parallel diodes model. Under reverse bias, the dominant leakage mechanism for the diodes affected by SELC has been identified as TAT with an extracted energy level of 0.2–0.3 eV. TEM imaging suggests that the cause of these electrical observations is defects in the AlGaN barrier formed during heavy-ion irradiation under bias. These defects are located near the anode edge. Potential avenues for future work would be to comprehensively investigate SELC in the complete transistor structure, as well as study the relationship between the angle of the heavy-ion strike and the resulting device damage.

SUPPLEMENTARY MATERIAL

See the supplementary material for the SEM image of the sacrificial characterization diode post-irradiation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Wiley Yu: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jeong-Kyu Kim**: Investigation (supporting); Writing – review & editing (supporting). **Srabanti Chowdhury**: Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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